

ECE 276B Project 3: Infinite-Horizon Stochastic Optimal Control

Winston Chou
 University of California, San Diego
 PID: A17460970
 La Jolla, CA, USA
 w3chou@ucsd.edu

Abstract—This report presents the implementation and evaluation of two optimal control strategies for trajectory tracking of a unicycle car model in a stochastic environment with obstacles. We first implement a receding-horizon Certainty Equivalent Control (CEC) scheme that solves a Non-Linear Programming (NLP) problem at each time step. Second, we utilize Generalized Policy Iteration (GPI) to solve the infinite-horizon stochastic optimal control problem through value iteration on a discretized state space. We compare the performance of both methods in tracking a complex lemniscate trajectory while maintaining safety. Our results highlight the trade-offs between online optimization (CEC) and offline value function pre-computation (GPI) regarding tracking accuracy and real-time feasibility.

Index Terms—Trajectory tracking, stochastic optimal control, Certainty Equivalent Control, Model Predictive Control, Generalized Policy Iteration, unicycle kinematics.

I. INTRODUCTION

Autonomous robot navigation requires precise trajectory tracking capabilities alongside robust obstacle avoidance. In many practical applications, the robot's dynamics are non-linear and subject to stochastic disturbances, while the environment contains complex constraints. This project focuses on the trajectory tracking problem for a unicycle robot tasked with following a periodic lemniscate path in the presence of circular obstacles and motion noise.

We explore two foundational approaches in optimal control:

- 1) **Certainty Equivalent Control (CEC)**: A receding-horizon approach that simplifies the stochastic control problem by solving a deterministic optimization problem at each step. By ignoring noise and assuming the most likely transition, CEC leverages efficient NLP solvers to find locally optimal control sequences.
- 2) **Generalized Policy Iteration (GPI)**: A dynamic programming-based approach that addresses the stochasticity directly. By discretizing the error-state space, GPI computes a value function and an optimal policy offline that minimizes the expected infinite-horizon discounted cost.

This report details the mathematical formulation of the unicycle dynamics and the tracking objective, describes the implementation of both algorithms, and provides a comparative analysis of their tracking performance and computational characteristics.

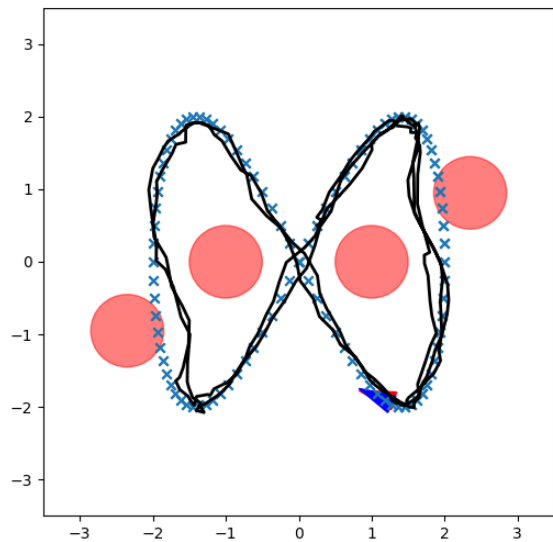


Fig. 1: Simulation of the unicycle car tracking a lemniscate trajectory in an environment with obstacles using Certainty Equivalent Control (CEC).

II. PROBLEM STATEMENT

The robot is modeled as a unicycle car operating in a 2D environment $\mathcal{W} \subset \mathbb{R}^2$ with state $x_t \in \mathbb{R}^3$ and control $u_t \in \mathcal{U} \subset \mathbb{R}^2$.

A. Robot Dynamics

The continuous-time kinematics are given by:

$$\dot{x} = \begin{bmatrix} \dot{p}_x \\ \dot{p}_y \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} v \cos \theta \\ v \sin \theta \\ \omega \end{bmatrix} \quad (1)$$

where v and ω are the linear and angular velocities. The discrete-time stochastic dynamics used in this project assume exact integration over a time step Δt :

$$x_{k+1} = f(x_k, u_k) + w_k, \quad w_k \sim \mathcal{N}(0, \Sigma) \quad (2)$$

where w_k represents the process noise.

B. Objective Function

The goal is to follow a reference trajectory $x_{ref}(t)$. We define the tracking error as $e_k = x_k - x_{ref}(k\Delta t)$. The objective is to minimize the expected infinite-horizon discounted cost:

$$J(x_0) = \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma^k \ell(e_k, u_k) \right] \quad (3)$$

where $\gamma \in (0, 1)$ is the discount factor and $\ell(e, u)$ is the stage cost:

$$\ell(e, u) = e_{pos}^T Q e_{pos} + q(1 - \cos e_\theta)^2 + u^T R u \quad (4)$$

Here $e_{pos} = [e_x, e_y]^T$, while Q , R , and q are weighting parameters.

C. Constraints

The control inputs are constrained within $[v_{min}, v_{max}]$ and $[\omega_{min}, \omega_{max}]$. Additionally, the robot must avoid a set of n circular obstacles $\mathcal{O} = \{O_j\}_{j=1}^n$ and remain within the world boundaries. Safety is enforced by ensuring the distance to any obstacle center $p_{O,j}$ satisfies:

$$\|p_x - p_{O,j}\|_2 \geq r_j + r_{robot} \quad (5)$$

where r_j is the obstacle radius and r_{robot} is the robot's collision margin.

III. TECHNICAL APPROACH

The trajectory tracking problem is addressed using two distinct paradigms: receding-horizon optimization and infinite-horizon value iteration.

A. Certainty Equivalent Control (CEC)

CEC simplifies the stochastic optimal control problem by assuming that process noise is zero during the planning horizon. At each time step k , we solve a finite-horizon deterministic optimal control problem:

$$\min_{u_{k:k+T-1}} \sum_{i=k}^{k+T-1} \gamma^{i-k} \ell(e_i, u_i) + \gamma^T \ell_{term}(e_{k+T}) \quad (6)$$

subject to the unicycle dynamics and control/state constraints. We implement this using the CasADi framework with the *ipopt* solver. Only the first control u_k^* is applied, and the optimization is repeated at the next time step.

B. Generalized Policy Iteration (GPI)

GPI addresses the infinite-horizon problem by pre-computing a stationary optimal policy $\pi^*(e)$. The continuous error-state space $\mathcal{E}$ is discretized into a finite grid $\mathcal{S}_{grid}$. We perform two interleaved steps:

- 1) **Policy Evaluation:** Estimate the value function V^π for the current policy π .
- 2) **Policy Improvement:** Update the policy by selecting the action that minimizes the one-step lookahead cost:

$$\pi'(e) = \arg \min_{u \in \mathcal{U}} \left\{ \ell(e, u) + \gamma \sum_{e'} P(e'|e, u) V^\pi(e') \right\} \quad (7)$$

To scale to large grids, we utilize Numba-accelerated JIT compilation for the transition matrix and cost computations, and enforce periodic wrap-around ($V_T = V_0$) for the lemniscate trajectory.

IV. RESULTS

A. CEC Performance

The CEC controller demonstrates robust tracking of the lemniscate trajectory in deterministic settings. However, as process noise increases, the receding-horizon solution may drift if the horizon T is too short.

B. GPI Performance

The GPI policy provides a fast lookup-based control law. Its performance is heavily dependent on the discretization resolution of the error states. We analyze the tracking error convergence and the impact of the discount factor γ .

C. Comparative Analysis

We compare CEC and GPI across several metrics:

- **Computational Cost:** Online optimization time (CEC) vs. offline pre-computation time and online lookup time (GPI).
- **Tracking Error:** Cumulative translation and rotation error.

V. CONCLUSION

In this project, we implemented and compared CEC and GPI for trajectory tracking of a unicycle robot. CEC offers flexibility in handling obstacle constraints through online optimization, while GPI provides a computationally efficient feedback law once the value function has converged.

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